

Equations of Tangent and Normal Lines

Numberbender | WORKSHEET



Name: _____ Date: _____ Score: _____

For each function and point, find the slope and equations of the tangent and normal lines.

Calculus 1 Worksheet #11 Equations of Tangent and Normal Lines

Learn: Tangent and Normal Curves

Remember: A derivative = slope of the Tangent line at that points x-coordinate

Example:

$$f(x) = x^2 + 3 \quad \text{pt}(1, 4), \quad f'(x) = 2x \Rightarrow f'(1) = 2$$

$$\text{Tangent line: } y - 4 = 2(x - 1) \Rightarrow y - 4 = 2x - 2 \Rightarrow y = 2x + 2$$

$$\text{Normal line: } y - 4 = -\frac{1}{2}(x - 1) \Rightarrow y - 4 = -\frac{1}{2}x + \frac{1}{2} \Rightarrow y = -\frac{1}{2}x + \frac{9}{2}$$

For the following:

1) Sketch a graph of $f(x)$. Use Graph Paper !!!!!

2) Find slope at point p.

3) Find equation of tangent at point p. Sketch line.

4) Find equation of normal at point p. Sketch line.

1. $y = x^2 - 3$	p(2, 1)	2. $y = 6 - x^2$	p(2, 2)
3. $y = 4x - x^2$	p(2, 4)	4. $y = x^2 - x - 6$	p(3, 0)
5. $y = x^3 - x$	p(1, 0)	6. $y = x^{1/2}$	p(4, 2)
7. $y = 6x^{-1}$	p(3, 2)	8. $y = x^3 - x$	p(-1, 0)
9. $y = 2 - \sqrt{x}$	p(4, 0)	10. $y = 4x^2 - x^4$	p($\sqrt{2}$, 4)
11. $y = 2 - 4x^{-2}$	p(2, 1)	12. $y = 1 + x^{2/3}$	p(0, 1)

13 – 17, at the specified point, find the equation of the normal to the curve $f(x)$.

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| 13. $f(x) = x^2$; (-3, _) | 14. $f(x) = x^2 - x$; (1, _) |
| 15. $f(x) = x^3$; (1, _) | 16. $f(x) = x^{1/2}$, $x \geq 0$; (1, _) |
| 17. $f(x) = 9x^{-1}$, $x \neq 0$; (3, _) | |

18. Use the **DEFINITION OF THE DERIVATIVE** to find $f'(x)$ if $f(x) = x^3 + 2x$



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Answer key — for instructor use only.

Answers:

1. $4, y = 4x - 7, y = -\frac{1}{4}x + \frac{3}{2}$	2. $-4, y = -4x + 10, y = \frac{1}{4}x + \frac{3}{2}$
3. $0, y = 4, x = 2$	4. $5, y = 5x - 15, y = -\frac{1}{5}x + \frac{3}{5}$
5. $2, y = 2x - 2, y = -\frac{1}{2}x + \frac{1}{2}$	6. $\frac{1}{4}, y = \frac{1}{4}x + 1, y = -4x + 18$
7. $-\frac{2}{3}, y = -\frac{2}{3}x + 4, y = \frac{3}{2}x - \frac{5}{2}$	8. $2, y = 2x + 2, y = -\frac{1}{2}x - \frac{1}{2}$
9. $-\frac{1}{4}, y = -\frac{1}{4}x + 1, y = 4x - 16$	10. $0, y = 4, x = \sqrt{2}$
11. $1, y = x - 1, y = -x + 3$	12. $\{ \}, \text{none}, \text{none}$
13. $y = \frac{1}{6}x + \frac{19}{2}$	14. $y = -x + 1$
15. $y = \frac{-1}{3}x + \frac{4}{3}$	16. $y = -2x + 3$
17. $y = x$	18. $3x^2 + 2$

